

3 An operon is a section of DNA found in prokaryotes.

(a) Explain why the enzymes coded for by the *lac* operon are described as inducible enzymes.

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..... [2]

(b) An investigation into the induction and action of the *lac* operon was carried out using the bacterium, *Escherichia coli*, grown in a growth medium containing glucose.

When the bacteria had used all the glucose, an excess of lactose was added to the growth medium. The activity of β -galactosidase was measured from this time (0 min), as shown in Fig. 3.1.

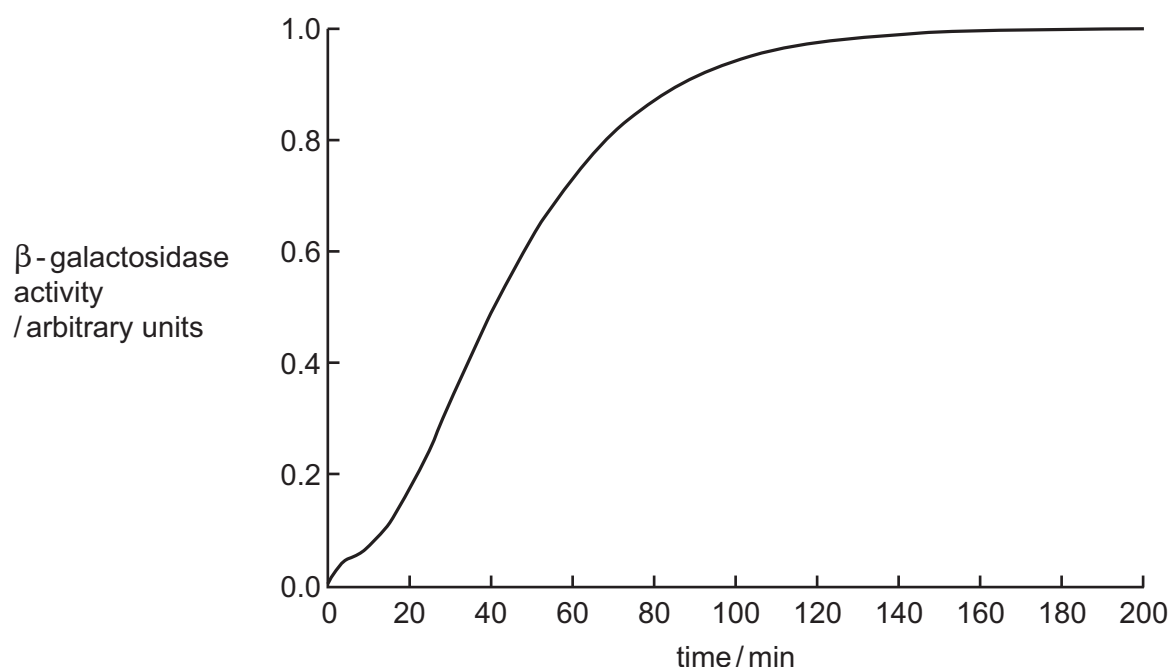


Fig. 3.1



[Total: 9]